# Assignment 1

**Question 1.** Solve each of the following systems.

(a)

$$\begin{aligned} 2x_1 + x_2 &= \text{-}2 \\ 4x_1 - 3x_2 &= 6 \end{aligned}$$

(b)

$$\begin{aligned} 4x_1 + 3x_2 &= 4 \\ \tfrac{2}{3}x_1 + 4x_2 &= 3 \end{aligned}$$

(c)

$$\begin{aligned} x_2 + x_3 + x_4 &= 0 \\ 3x_1 + 3x_3 - 4x_4 &= 7 \\ x_1 + x_2 + x_3 + 2x_4 &= 6 \\ 2x_1 + 3x_2 + x_3 + 3x_4 &= 6 \end{aligned}$$

**Solution**

(a) (0, -2)
(b) (1/2, 2/3)
(c) (4, -3, 1, 2)

**Question 2.** Write out the system of equations that corresponds to the following augmented matrix:

$$\left(\begin{array}{ccc|c} 5 & 3 & 1 & 3 \\ 2 & 3 & -4 & 0 \end{array}\right)$$

**Solution**

$$5x_1 + 3x_2 + x_3 = 3$$
$$2x_1 + 3x_2 - 4x_3 = 0$$

**Question 3.** Consider a system of the form

$$a_{11}x_1 + a_{12}x_2 = 0$$
$$a_{21}x_1 + a_{22}x_2 = 0$$

where $a_{11}, a_{12}, a_{21}$, and $a_{22}$ are constants. Explain why a system of this form must be consistent.

**Solution**

When $a_{11} \neq 0$, the augmented matrix for the system is

$$\left(\begin{array}{cc|c} a_{11} & a_{12} & 0 \\ a_{21} & a_{22} & 0 \end{array}\right) \xrightarrow{R_2 \to R_2 - \frac{a_{21}}{a_{11}} R_1} \left(\begin{array}{cc|c} a_{11} & a_{12} & 0 \\ 0 & \frac{a_{22}a_{11} - a_{12}a_{21}}{a_{11}} & 0 \end{array}\right)$$

If $a_{22}a_{11} - a_{12}a_{21} = 0$, $x_2$ will be free variable, then the system is consistent.

If $a_{22}a_{11}a_{12}a_{21} \neq 0$, the system will have a unique solution (0, 0).

Obviously, the results above can be expanded to the situation that at least one of the four constants $\neq 0$.
And when all four constants $= 0$, the system will have infinite solutions.
So a system of this form must be consistent.

**Question 4.** Given a system of the form

$$-m_1x_1 + x_2 = b_1$$
$$-m_2x_1 + x_2 = b_2$$

where $m_1, m_2, b_1$, and $b_2$ are constants:
(a) Show that the system will have a unique solution if $m_1 \neq m_2$.
(b) Show that if $m_1 = m_2$, then the system will be consistent only if $b_1 = b_2$.
(c) Give a geometric interpretation of parts (a) and (b).

**Solution**

(a) When $m_1 = 0, m_2 \neq 0$, the augmented matrix for the system is

$$\left(\begin{array}{cc|c} 0 & 1 & b_1 \\ -m_2 & 1 & b_2 \end{array}\right)$$

By substitution, we could obtain a unique solution $\left(\frac{b_1-b_2}{m_2}, b_1\right)$.
When $m_2 = 0, m_1 \neq 0$, the augmented matrix for the system is

$$\left(\begin{array}{cc|c} -m_1 & 1 & b_1 \\ 0 & 1 & b_2 \end{array}\right)$$

By substitution, we could obtain a unique solution $\left(\frac{b_2-b_1}{m_1}, b_2\right)$.
When $m_1 \neq 0, m_2 \neq 0$, the augmented matrix for the system is

$$\left(\begin{array}{cc|c} -m_1 & 1 & b_1 \\ -m_2 & 1 & b_2 \end{array}\right) \overset{R_2 \to R_2 - \frac{m_2}{m_1}R_1}{\longrightarrow} \left(\begin{array}{cc|c} -m_1 & 1 & b_1 \\ 0 & 1-\frac{m_2}{m_1} & b_2 - \frac{b_1m_2}{m_1} \end{array}\right)$$

By substitution, we could obtain a unique solution $\left(\frac{b_2-b_1}{m_1-m_2}, \frac{m_1b_2-b_1m_2}{m_1-m_2}\right)$.
(b) When $m_1 = m_2$, the augmented matrix for the system is

$$\left(\begin{array}{cc|c} -m_1 & 1 & b_1 \\ -m_2 & 1 & b_2 \end{array}\right) \overset{R_2 \to R_2 - R_1}{\longrightarrow} \left(\begin{array}{cc|c} -m_1 & 1 & b_1 \\ 0 & 0 & b_2 - b_1 \end{array}\right)$$

So only if $b_2 - b_1 = 0$, that is $b_1 = b_2$, the system will be consistent.

(c) For part (a):
If $m_1 \neq m_2$, the two lines will intersect at one point, that is a unique solution.

For part(b) :
If $b_1 \neq b_2$, the two lines will be parallel, that is no solution.
Only if $b_1 = b_2$, both equations will represent the same line, that is infinite solutions.

**Question 5.** Which of the matrices that follow are in the row echelon form? Which are in the reduced row echelon form?

(a)
$$\begin{pmatrix} 1 & 2 & 0 & 4 \\ 0 & 0 & 1 & 2 \end{pmatrix}$$

(b)
$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(c)
$$\begin{pmatrix} 1 & 3 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

(d)
$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

(e)
$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{pmatrix}$$

(f)
$$\begin{pmatrix} 1 & 4 & 6 \\ 0 & 0 & 1 \\ 0 & 1 & 3 \end{pmatrix}$$

(g)
$$\begin{pmatrix} 1 & 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & 2 & 4 \\ 0 & 0 & 1 & 3 & 6 \end{pmatrix}$$

(h)

$$\begin{pmatrix} 0 & 1 & 3 & 4 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

**Solution**

Row echelon form: (a), (c), (d), (e), (g), (h).

Reduced row echelon form: (a), (c), (d), (g).

**Question 6.** For each of the following systems, use Gaussian elimination to obtain an equivalent system whose coefficient matrix is in the row echelon form. Indicate whether the system is consistent. Compute the solution set if the system is consistent.

(a)

$$x_1 - 2x_2 = 3$$
$$2x_1 - x_2 = 9$$

(b)

$$3x_1 + 2x_2 - x_3 = 4$$
$$x_1 - 2x_2 + 2x_3 = 1$$
$$11x_1 + 2x_2 + x_3 = 14$$

(c)

$$x_1 + x_2 + x_3 + x_4 = 0$$
$$2x_1 + 3x_2 - x_3 - x_4 = 2$$
$$3x_1 + 2x_2 + x_3 + x_4 = 5$$
$$3x_1 + 6x_2 - x_3 - x_4 = 4$$

**Solution**

(a)

$$\left(\begin{array}{cc|c} 1 & -2 & 3 \\ 0 & 1 & 1 \end{array}\right)$$

The system is consistent, the solution is $(5, 1)$.

(b)

$$\left(\begin{array}{ccc|c} 1 & -2 & 2 & 1 \\ 0 & 1 & -\frac{7}{8} & \frac{1}{8} \\ 0 & 0 & 0 & 0 \end{array}\right)$$

The system is consistent, the solution is $\left(\frac{5-x_3}{4}, \frac{1+7x_3}{8}, x_3\right)$.

(c)

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & -3 & -3 & 0 \\ 0 & 0 & 1 & 1 & -\frac{2}{5} \\ 0 & 0 & 0 & 0 & -\frac{21}{5} \end{array}\right)$$

The system is not consistent.

**Question 7.** The augmented matrices that follow are in the row echelon form. For each case, indicate whether the corresponding linear system is consistent. If the system has a unique solution, find it.

(a)

$$\left(\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{array}\right)$$

(b)

$$\left(\begin{array}{cc|c} 1 & 3 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{array}\right)$$

(c)

$$\left(\begin{array}{ccc|c} 1 & -2 & 4 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

(d)

$$\left(\begin{array}{ccc|c} 1 & -2 & 2 & -2 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & 2 \end{array}\right)$$

(e)

$$\left(\begin{array}{ccc|c} 1 & 3 & 2 & -2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{array}\right)$$

**Solution**

(a) Inconsistent;

(b) consistent, (4,-1);

(c) consistent, infinitely many solutions;

(d) consistent, (4, 5, 2);

(e) Inconsistent.

**Question 8.** Consider a linear system whose augmented matrix is of the form

$$\left[\begin{array}{rrr|r} 1 & 2 & 1 & 1 \\ -1 & 4 & 3 & 2 \\ 2 & -2 & a & 3 \end{array}\right]$$

For what values of $a$ will the system have a unique solution?

**Solution**

The REF can be obtained from the given augmented matrix by elementary row operations as

$$\left[\begin{array}{rrr|r} 1 & 2 & 1 & 1 \\ 0 & 6 & 4 & 3 \\ 0 & 0 & a+2 & 4 \end{array}\right]$$

to make the system have an unique solution, $a + 2 \neq 0 \Rightarrow a \neq -2$.

**Question 9.** The augmented matrices that follow are in the reduced row echelon form. In each case, find the solution set to the corresponding linear system.

(a)

$$\left(\begin{array}{rrr|r} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

(b)

$$\left(\begin{array}{rrr|r} 1 & -3 & 0 & 2 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

(c)

$$\left(\begin{array}{rrrr|r} 1 & 5 & -2 & 0 & 3 \\ 0 & 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

**Solution**

(a) The solution is $(-2, 5, 3)$.

(b) The solution is $\{(2 + 3\alpha, \alpha, \text{-2}) \mid \alpha \text{ real } \}$.

(c) The solution is $\{(3 - 5\alpha + 2\beta, \alpha, \beta, 6) \mid \alpha, \beta \text{ real } \}$.

**Question 10.** Use Gauss-Jordan reduction to solve each of the following systems.

(a)

$$\begin{aligned} x_1 + 3x_2 + x_3 + x_4 &= 3 \\ 2x_1 - 2x_2 + x_3 + 2x_4 &= 8 \\ 3x_1 + x_2 + 2x_3 - x_4 &= -1 \\ x_1 + x_2 + x_3 + x_4 &= 0 \end{aligned}$$

(b)

$$\begin{aligned} 2x_1 + x_2 - x_3 + 3x_4 &= 0 \\ x_1 - 2x_2 + x_3 + x_4 &= 0 \end{aligned}$$

**Solution**

(a) The solution is $\left\{\left(\frac{3}{4} - \frac{5}{8}\alpha, -\frac{1}{4} - \frac{1}{8}\alpha, \alpha, 3\right) \mid \alpha \text{ real } \right\}$.
(b) The solution is $\left\{\alpha\left(-\frac{4}{3}, 0, \frac{1}{3}, 1\right) \mid \alpha \text{ real } \right\}$.

**Question 11.** For the following system of the linear equations

$$\begin{cases} 4x_1 + 5x_2 + 3x_3 + 3x_4 + 4x_5 = -5 \\ 2x_1 + 3x_2 + x_3 + x_5 = -3 \\ 3x_1 + 4x_2 + 2x_3 + x_4 + x_5 = -1 \end{cases}$$

1) Write down the coefficient matrix and the augmented matrix
2) Solve the system by Gauss-Jordan elimination

**Solution**

1) coefficient matrix: $\begin{bmatrix} 4 & 5 & 3 & 3 & 4 \\ 2 & 3 & 1 & 0 & 1 \\ 3 & 4 & 2 & 1 & 1 \end{bmatrix}$,

augmented matrix: $\left[\begin{array}{ccccc|c} 4 & 5 & 3 & 3 & 4 & -5 \\ 2 & 3 & 1 & 0 & 1 & -3 \\ 3 & 4 & 2 & 1 & 1 & -1 \end{array}\right]$

2)

$$\left[\begin{array}{ccccc|c} 4 & 5 & 3 & 3 & 4 & -5 \\ 2 & 3 & 1 & 0 & 1 & -3 \\ 3 & 4 & 2 & 1 & 1 & -1 \end{array}\right] \Rightarrow \left[\begin{array}{ccccc|c} 4 & 5 & 3 & 3 & 4 & -5 \\ 0 & \frac{1}{2} & -\frac{1}{2} & -\frac{3}{2} & -1 & -\frac{1}{2} \\ 0 & \frac{1}{4} & -\frac{1}{4} & -\frac{5}{4} & -2 & \frac{11}{4} \end{array}\right]$$

$$\Rightarrow \left[\begin{array}{ccccc|c} 4 & 5 & 3 & 3 & 4 & -5 \\ 0 & \frac{1}{2} & -\frac{1}{2} & -\frac{3}{2} & -1 & -\frac{1}{2} \\ 0 & 0 & 0 & -\frac{1}{2} & -\frac{3}{2} & 3 \end{array}\right]$$

$$\Rightarrow \left[\begin{array}{ccccc|c} 4 & 5 & 3 & 3 & 4 & -5 \\ 0 & 1 & -1 & -3 & -2 & -1 \\ 0 & 0 & 0 & 1 & 3 & -6 \end{array}\right] \Rightarrow \left[\begin{array}{ccccc|c} 1 & 0 & 2 & 0 & -10 & 27 \\ 0 & 1 & -1 & 0 & 7 & -19 \\ 0 & 0 & 0 & 1 & 3 & -6 \end{array}\right]$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} s + \begin{bmatrix} 10 \\ -7 \\ 0 \\ -3 \\ 1 \end{bmatrix} t + \begin{bmatrix} 27 \\ -19 \\ 0 \\ -6 \\ 0 \end{bmatrix}, s \in R, t \in R$$

**Question 12.**

Prove that the reduced row echelon form of any matrix $A \in \mathbb{R}^{m \times n}$ is unique.

**Solution**

Proof. Let $A$ be a matrix and suppose it has two row reduced echelon forms say $B$ and $C$. That means applying a sequence of row operations to $A$ we got $B$ and applying another sequence of row operations we got $C$. We need to show that $B = C$.

Note that $A, B, C$ are row equivalent to each other since row operation gives a row equivalent matrix. That means every row in $A$ is a linear combination of rows of $B$ and vice versa. Similarly every row in $A$ is a linear combination of rows of $C$ and vice versa.

On the contrary let us assume that $B$ and $C$ are not equal. Then select the leftmost column where they differ and also select all pivot columns (leading 1 columns) to the left of this column giving rise to two matrices say $R$ and $S$. Since $B$ and $C$ were row rquivalent the matrices $R$ and $S$ are row equivalent since deletion of columns does not affect row equivalence.

Note that after interchanging some rows (if requird) the matrices $R$ and $S$ look like:

$$R = \left[\begin{array}{c|c} I_{r\times r} & \mathbf{r} \\ \hline 0 & 0 \end{array}\right] S = \left[\begin{array}{c|c} I_{r\times r} & \mathbf{s} \\ \hline 0 & 0 \end{array}\right]$$

It follows that $R$ and $S$ are row equivalent since deletion of columns (variables simultaneously) does not affect row equivalence, and that they are reduced but not equal. Now we treat these matrices as augmented matrices of two linear systems. The system for $R$ has a unique solution $\mathbf{r}$ or is inconsistent, respectively. Similarly, the system for $S$ has a unique solution s or is incon-

sistent, respectively. Since the systems are equivalent, $r = s$ or both systems are inconsistent. Either way we have $R = S$, a contradiction.

**Question 13.**

Find non-zero $2 \times 2$ matrices $A$ and $B$ such that $AB = O$.

**Solution**

Consider, $A = \begin{bmatrix} 1 & -1 \\ 3 & -3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$

$$AB = \begin{bmatrix} 1 & -1 \\ 3 & -3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 \times 1 + (-1) \times 1 & 1 \times 2 + (-1) \times 2 \\ 3 \times 1 + (-3) \times 1 & 3 \times 2 + (-3) \times 2 \end{bmatrix}$$

$$AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$$

**Question 14.** Consider a linear system whose augmented matrix is of the form

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 1 & 3 & a & b \end{array}\right]$$

(a) For what values of $a$ and $b$ will the system have infinitely many solutions?

(b) For what values of $a$ and $b$ will the system be inconsistent?

**Solution**

(a) A row-echelon form for the system is

$$\begin{bmatrix} 1 & 1 & 3 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & a-5 & b-4 \end{bmatrix}$$

There will be infinitely many solutions if there is a free variable, and this can only happen here is the the bottom row is $\begin{bmatrix} 0 & 0 & u & v \end{bmatrix}$, where $u = 0$ and $v = 0$. Thus, $a = 5$ and $b = 4$.

(b) This can only happen here is the the bottom row is $\begin{bmatrix} 0 & 0 & u & v \end{bmatrix}$, where $u = 0$ and $v \neq 0$. Thus, $a = 5$ and $b \neq 4$.

**Question 15.** Let

$$\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Compute $A^2$ and $A^3$. What will $A^n$ turn out to be? Note that the rigorous proof is required for the full score.

**Solution**

$$A^2 = AA = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} = A$$

$$A^3 = A^2A = AA = A$$

Thus, we can prove that for any $n > 2$

$$A^n = A^{n-2}A^2 = A^{n-2}A = A^{n-1}$$

Iteratively and with all forementioned results, it holds true that for any $n > 0$,

$$A^n = A^{n-1} = \ldots = A^3 = A^2 = A$$

**Question 16.** Let

$$A = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Compute $A^2$ and $A^3$. What will $A^{2n}$ and $A^{2n+1}$ turn out to be? Note that the rigorous proof is required for the full score.

**Solution**

$$A^2 = AA = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= I$$

$$A^3 = A^2A = IA = A$$

Thus, iteratively,we can prove that for any $n > 1$,

$$A^{2n} = A^{2n-2}A^2 = A^{2n-2}I = A^{2(n-1)}$$

Iteratively and with all forementioned results, it holds true that for any $n > 0$,

$$A^{2n} = A^{2(n-1)} = \ldots = A^{2\times 2} = A^{2\times 1} = I$$

Thus, for any $n > 0$,

$$A^{2n+1} = A^{2n}A = IA = A$$